

Se-Sp_analysis

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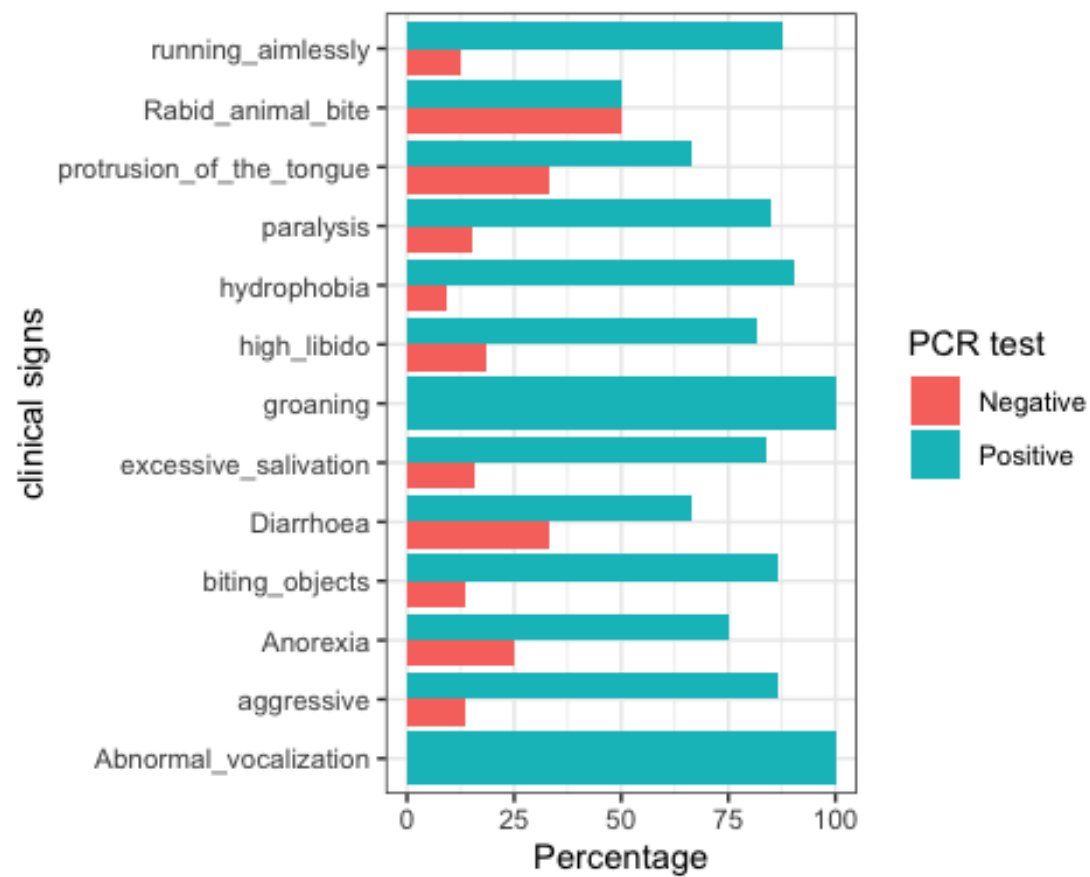
2024-09-03

Se, Sp, NPV, PPV

```
##           Outcome +   Outcome -   Total
## Test +           90           6      96
## Test -           14           2      16
## Total           104           8     112
##
## Point estimates and 95% CIs:
## -----
## Apparent prevalence *           0.86 (0.78, 0.92)
## True prevalence *             0.93 (0.86, 0.97)
## Sensitivity *                 0.87 (0.78, 0.92)
## Specificity *                 0.25 (0.03, 0.65)
## Positive predictive value *    0.94 (0.87, 0.98)
## Negative predictive value *    0.12 (0.02, 0.38)
## Positive likelihood ratio      1.15 (0.77, 1.73)
## Negative likelihood ratio      0.54 (0.15, 1.97)
## False T+ proportion for true D- * 0.75 (0.35, 0.97)
## False T- proportion for true D+ * 0.13 (0.08, 0.22)
## False T+ proportion for T+ *    0.06 (0.02, 0.13)
## False T- proportion for T- *    0.88 (0.62, 0.98)
## Correctly classified proportion * 0.82 (0.74, 0.89)
## -----
## * Exact CIs
```

PCR +VE and -VE of the relevant variables

```
##
## Abnormal_vocalization aggressive Anorexia biting_objects Diarrhoea
## 0           0           10           3           8           1
## 1           4           65           9          52           2
##
## excessive_salivation groaning high_libido hydrophobia paralysis
## 0           10           0           5           3           5
## 1           53           2          22          29          28
##
## protrusion_of_the_tongue Rabid_animal_bite running_aimlessly
## 0           1           1           1           6
## 1           2           1           1          43
```



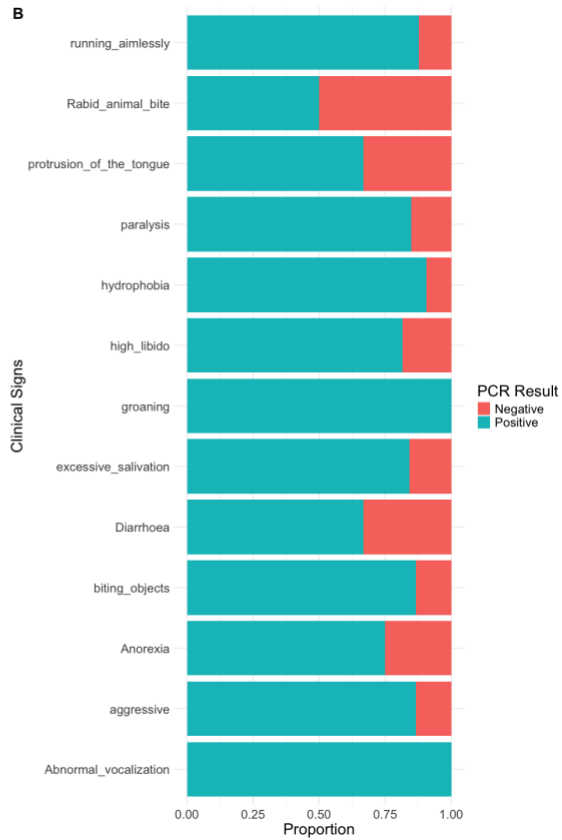
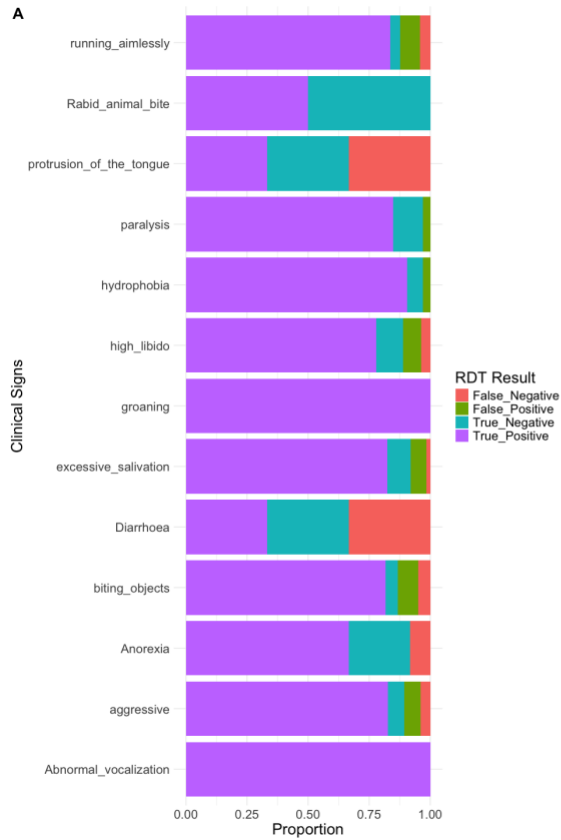


Table species vs the symptoms

species	Anorexia	Diarrhoea	aggressive	biting_objects	excessive_salivation	groaning	high_libido	hydrophobia	paralysis	protrusion_of_the_tongue	running_aimlessly	Abnormal_vocalization	Rabid_animal_bite
Dog	3	3	58	48	44	2	18	17	24	3	39	0	0
Goat	6	0	11	11	5	0	5	4	2	0	4	3	1
Cattle	3	0	5	0	13	0	3	10	7	0	5	1	0
Sheep	0	0	1	1	1	0	1	1	0	0	1	0	1
Total	12	3	75	60	63	2	27	32	33	3	49	4	2

GLM

```
reg_model <- table(test$clinical_signs_, test$test_result)

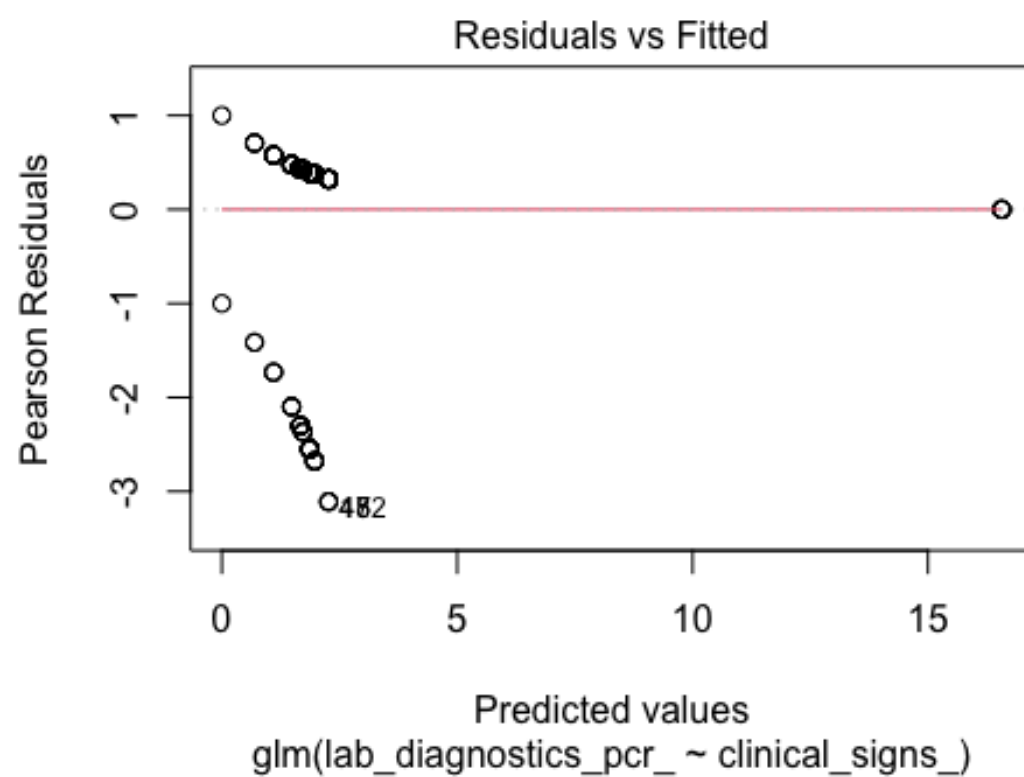
chisq.test(reg_model)

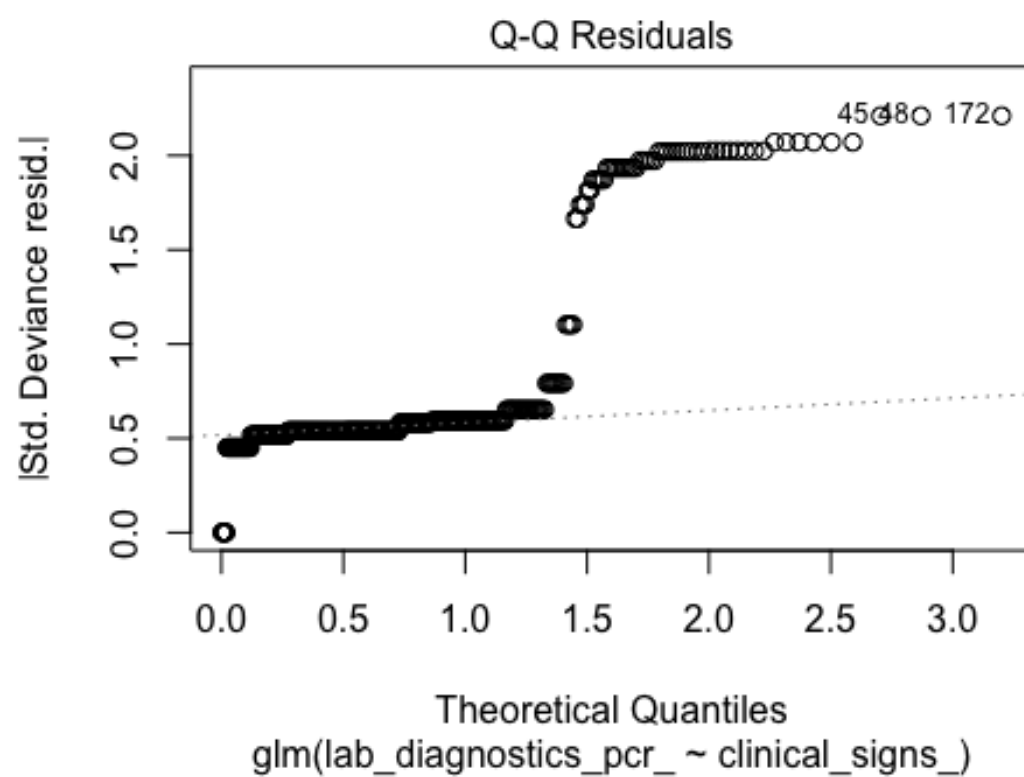
## Warning in stats::chisq.test(x, y, ...): Chi-squared approximation may be
## incorrect

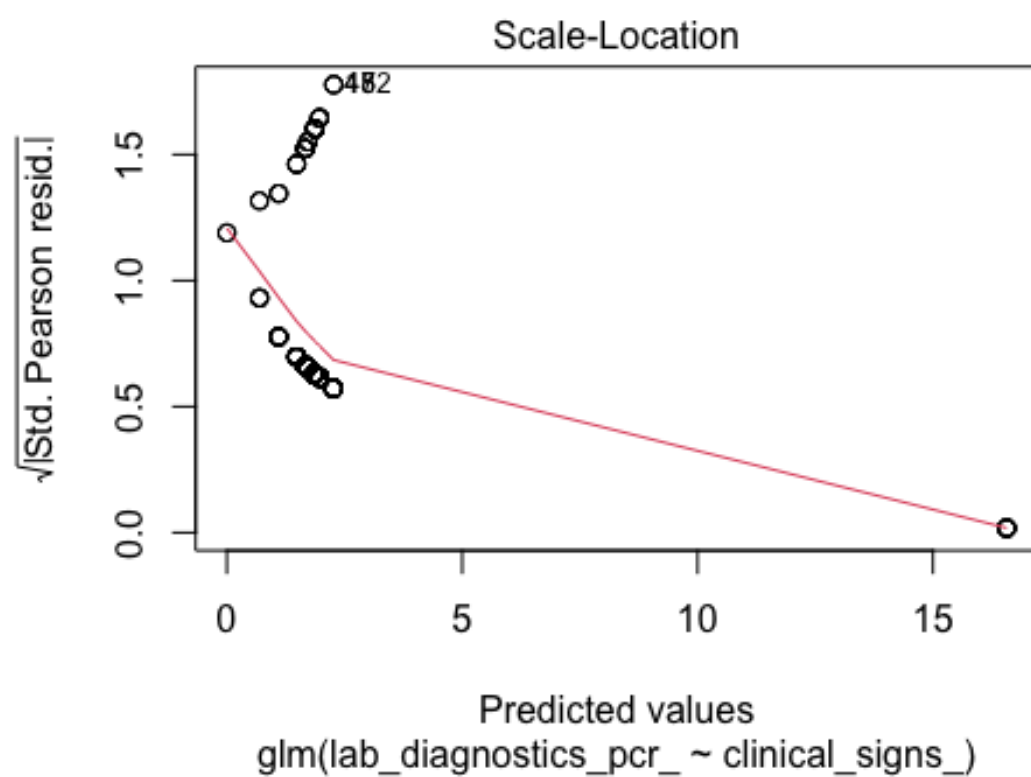
##
## Pearson's Chi-squared test
##
## data:  reg_model
## X-squared = 41.994, df = 36, p-value = 0.2272

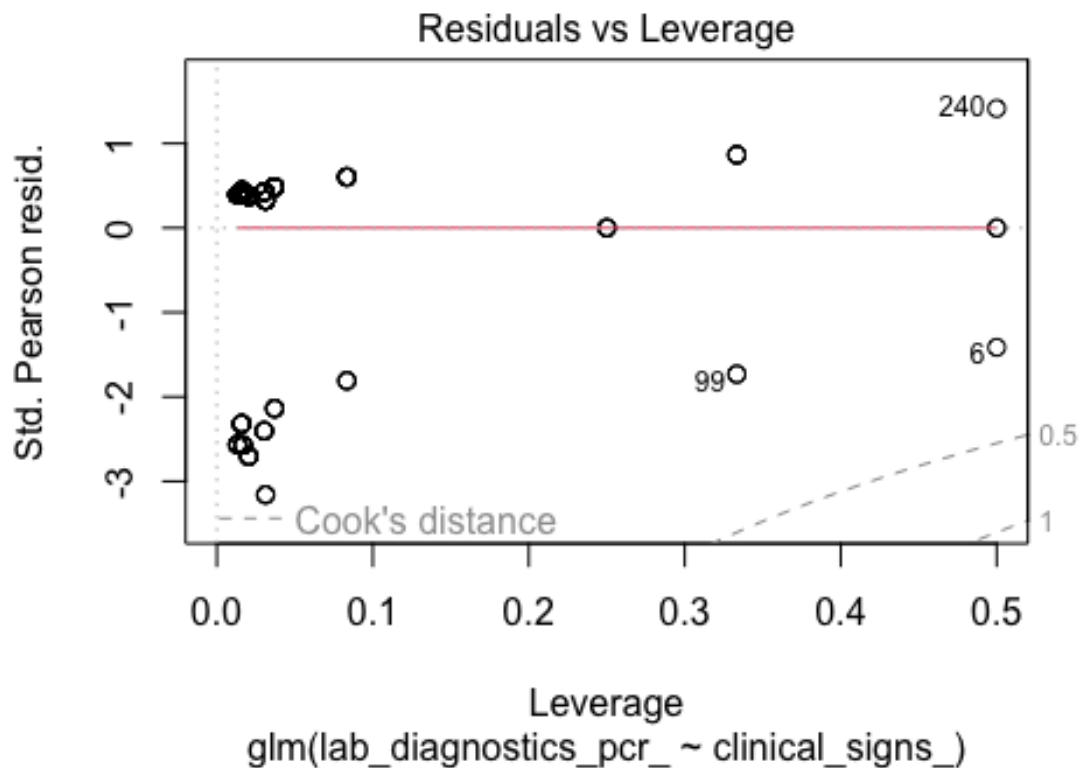
model1 <- glm(lab_diagnostics_pcr_ ~ clinical_signs_ , family = binomial(link
= 'logit'), data = test)

plot(model1) #To examine the residuals of the GLM which can help identify
outliers or patterns that may be affecting convergence.
```









```
summary(model1)
```

```
##
## Call:
## glm(formula = lab_diagnostics_pcr_ ~ clinical_signs_, family =
binomial(link = "logit"),
##   data = test)
##
## Coefficients:
##
##               Estimate Std. Error z value
Pr(>|z|)
## (Intercept)      1.657e+01  1.200e+03   0.014
0.989
## clinical_signs_aggressive      -1.469e+01  1.200e+03  -0.012
0.990
## clinical_signs_Anorexia      -1.547e+01  1.200e+03  -0.013
0.990
## clinical_signs_biting_objects      -1.469e+01  1.200e+03  -0.012
0.990
## clinical_signs_Diarrhoea      -1.587e+01  1.200e+03  -0.013
0.989
## clinical_signs_excessive_salivation      -1.490e+01  1.200e+03  -0.012
0.990
```

```

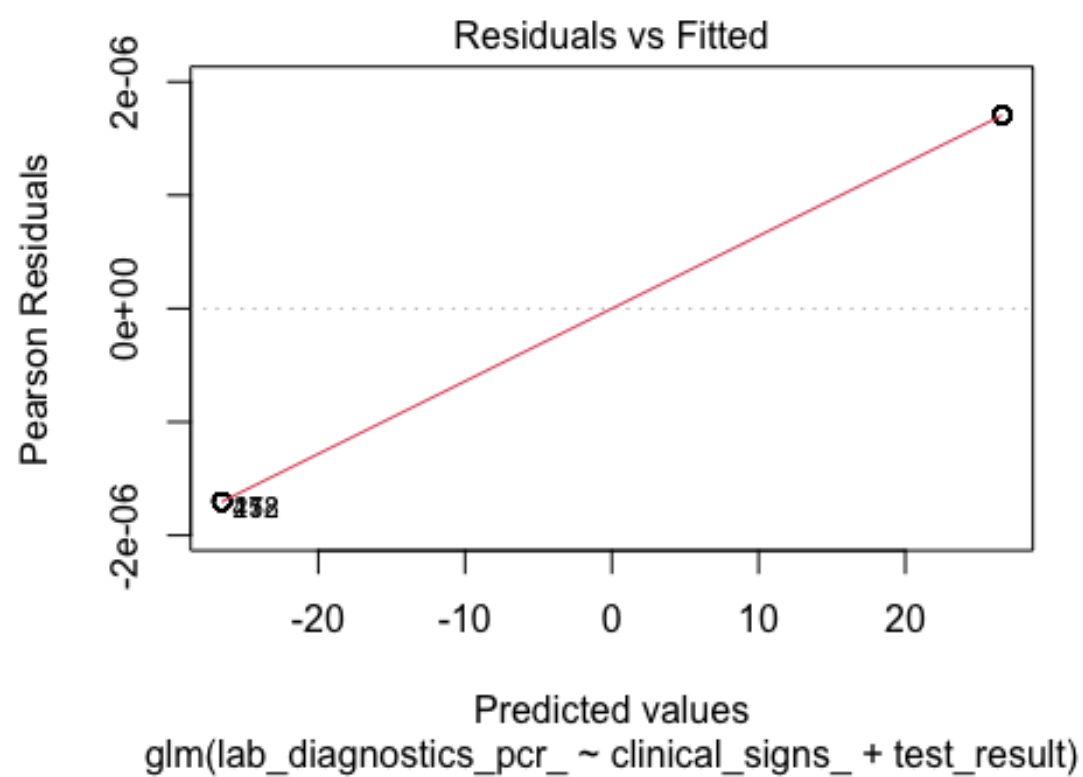
## clinical_signs_groaning          9.624e-09  2.078e+03  0.000
1.000
## clinical_signs_high_libido       -1.508e+01  1.200e+03 -0.013
0.990
## clinical_signs_hydrophobia       -1.430e+01  1.200e+03 -0.012
0.990
## clinical_signs_paralysis          -1.484e+01  1.200e+03 -0.012
0.990
## clinical_signs_protrusion_of_the_tongue -1.587e+01  1.200e+03 -0.013
0.989
## clinical_signs_Rabid_animal_bite  -1.657e+01  1.200e+03 -0.014
0.989
## clinical_signs_running_aimlessly  -1.460e+01  1.200e+03 -0.012
0.990
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 302.44  on 364  degrees of freedom
## Residual deviance: 295.35  on 352  degrees of freedom
## AIC: 321.35
##
## Number of Fisher Scoring iterations: 15

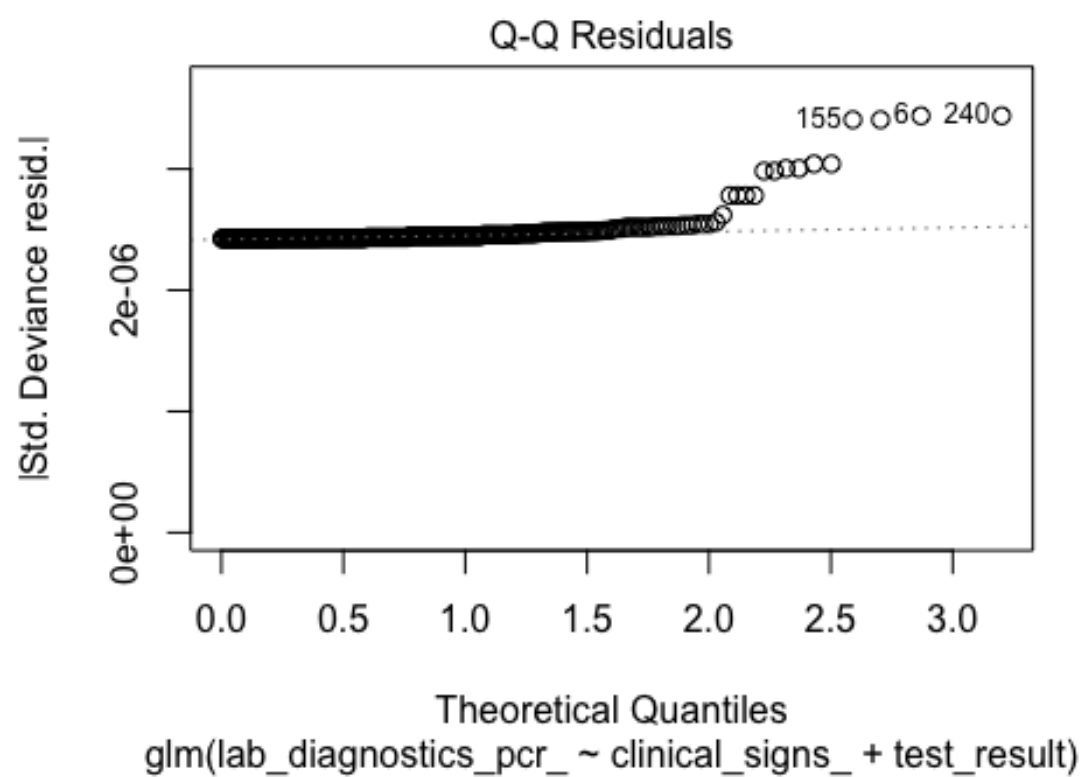
model2 <- glm(lab_diagnostics_pcr_ ~ clinical_signs_ + test_result, family =
binomial(link = 'logit'), data = test)

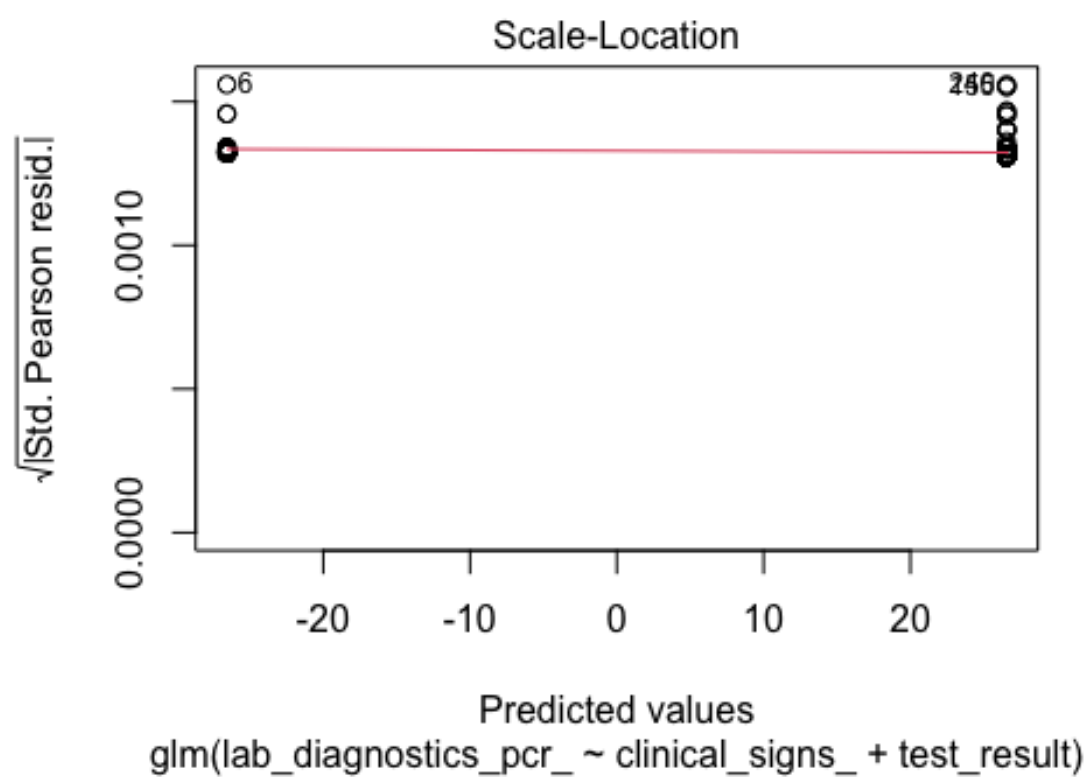
## Warning: glm.fit: algorithm did not converge

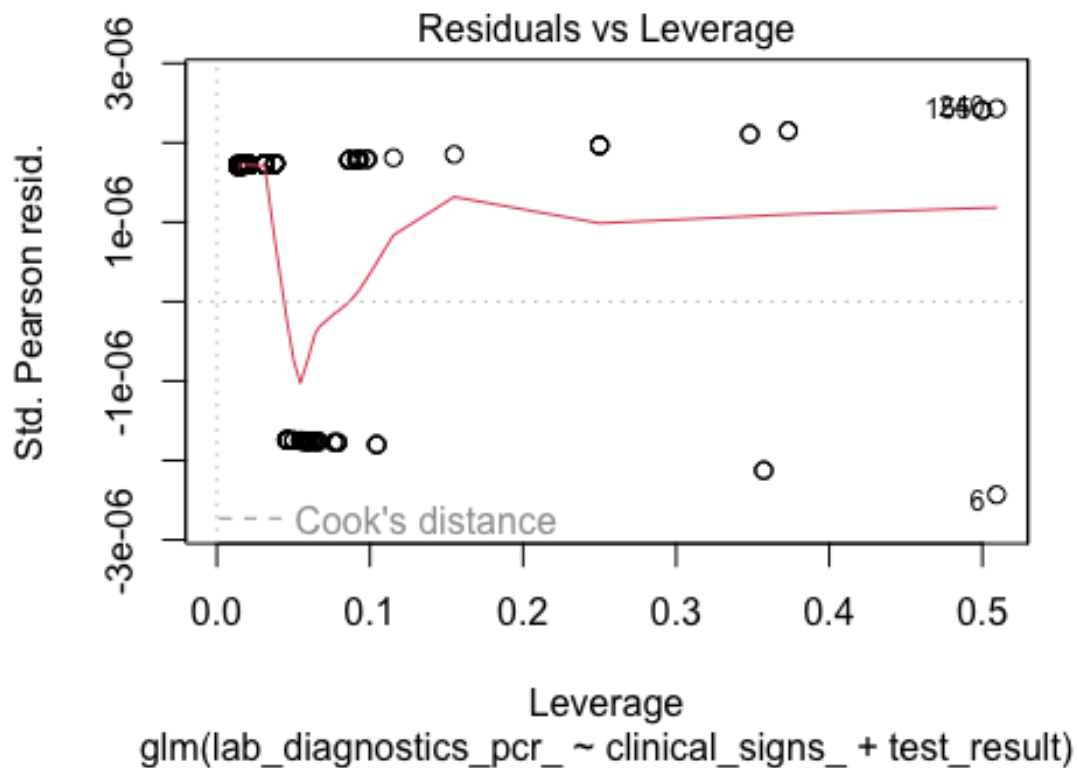
plot(model2)

```









```
summary(model2)
```

```
##
## Call:
## glm(formula = lab_diagnostics_pcr_ ~ clinical_signs_ + test_result,
##      family = binomial(link = "logit"), data = test)
##
## Coefficients:
##
##              Estimate Std. Error z value
Pr(>|z|)
## (Intercept)          2.657e+01  2.062e+05      0
1
## clinical_signs_aggressive -3.777e-06  1.830e+05      0
1
## clinical_signs_Anorexia  -6.964e-07  2.066e+05      0
1
## clinical_signs_biting_objects -4.464e-07  1.842e+05      0
1
## clinical_signs_Diarrhoea -2.076e-06  2.754e+05      0
1
## clinical_signs_excessive_salivation -4.987e-07  1.838e+05      0
1
## clinical_signs_groaning  -1.766e-07  3.084e+05      0
```

```

1
## clinical_signs_high_libido          -5.608e-07  1.911e+05    0
1
## clinical_signs_hydrophobia          -3.696e-07  1.889e+05    0
1
## clinical_signs_paralysis             -3.947e-07  1.888e+05    0
1
## clinical_signs_protrusion_of_the_tongue -2.076e-06  2.754e+05    0
1
## clinical_signs_Rabid_animal_bite     -2.062e-06  3.103e+05    0
1
## clinical_signs_running_aimlessly     -4.574e-07  1.854e+05    0
1
## test_resultFalse_Positive            -5.313e+01  1.272e+05    0
1
## test_resultTrue_Negative             -5.313e+01  1.195e+05    0
1
## test_resultTrue_Positive              3.499e-06  1.040e+05    0
1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 3.0244e+02  on 364  degrees of freedom
## Residual deviance: 2.1176e-09  on 349  degrees of freedom
## AIC: 32
##
## Number of Fisher Scoring iterations: 25

```